

Fake News Detection from Articles



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1 Topic and Motivation

1.1 Introduction

The topic of this project is **Fake News Detection from articles using Machine Learning**).

In recent years, fake news has become one of the biggest problems on the internet. Social media platforms and online news websites allow information to spread extremely quickly. Unfortunately, false or misleading information can also spread rapidly.

Fake news can influence:

- political opinions
- elections
- public health decisions
- financial markets
- social stability

For example, during elections or global health crises, misinformation can cause panic, confusion, and incorrect decision-making.

1.2 Problem Importance

Manual detection of fake news is extremely difficult because:

- millions of articles are published daily
- fake news often resembles real news
- humans cannot efficiently verify all information

Therefore, automated systems are necessary.

1.3 How Machine Learning Helps

Machine learning algorithms can automatically learn hidden patterns from textual data and classify news articles as fake or real.

Machine learning models analyze:

- writing style
- word usage
- emotional tone
- punctuation patterns
- sentence structure
- vocabulary distribution

The main goal of this project is to build a machine learning system capable of accurately detecting fake news articles.

2 Dataset Collection and Preparation

2.1 Dataset Sources

To create a reliable and robust model, multiple datasets were combined.

The datasets used in this project are:

- ISOT Fake News Dataset
- LIAR Dataset
- WELFake Dataset

The datasets were collected from:

- Kaggle
- public repositories
- research datasets

2.2 Why Multiple Datasets Were Used

Combining multiple datasets improves:

- data diversity
- model generalization
- robustness
- resistance to overfitting

Using multiple sources also reduces dataset-specific bias and creates a more realistic training environment.

2.3 Data Cleaning

The raw datasets contained:

- missing values
- duplicate articles
- noisy text
- URLs
- HTML tags
- inconsistent formatting

Several preprocessing steps were applied.

2.3.1 Missing Value Removal

Rows with missing text or labels were removed.

2.3.2 Duplicate Removal

Duplicate articles were removed to prevent model bias and data leakage.

2.3.3 Text Cleaning

The following cleaning operations were performed:

- lowercase conversion
- URL removal
- HTML tag removal
- punctuation cleaning
- removal of special characters

2.3.4 Stop Word Removal

Common words such as:

- the
- and
- is
- of

were removed because they do not provide meaningful information for classification.

2.3.5 Lemmatization

Words were converted into their base forms.

Examples:

- running → run
- studies → study
- better → good

Lemmatization improves consistency and reduces vocabulary size.

2.4 Feature Engineering

Additional stylometric features were created to capture writing style differences between fake and real news.

The engineered features include:

- text length
- word count
- average word length
- capitalization ratio
- number of exclamation marks
- number of question marks

These features help identify emotional or sensational writing patterns commonly found in fake news.

3 Research and Exploratory Data Analysis (EDA)

3.1 Related Research

Several research papers and articles related to fake news detection were reviewed.

The literature mainly focused on:

- NLP techniques
- TF-IDF vectorization
- stylometric analysis
- text classification models
- misinformation detection

Previous studies show that classical machine learning models such as Logistic Regression and Support Vector Machines perform very well on text classification tasks.

3.2 Exploratory Data Analysis

EDA was performed to better understand the dataset and identify patterns between fake and real news.

3.2.1 Class Distribution

The distribution of fake and real news articles was analyzed to check dataset balance.

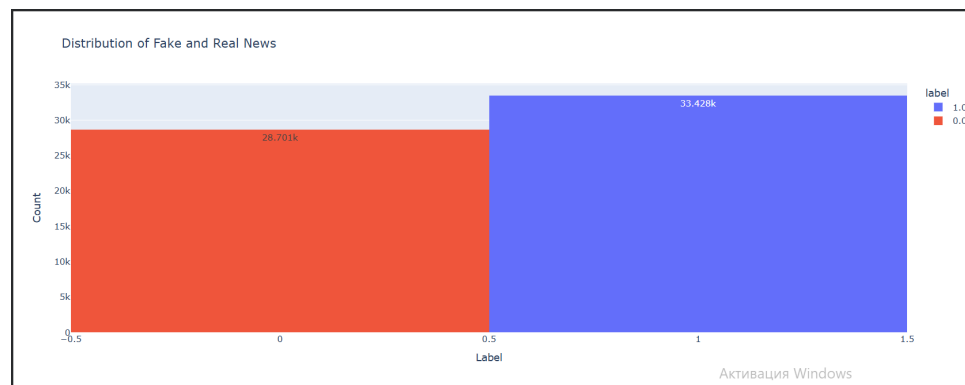


Figure 1: Distribution of Fake and Real News

The dataset is relatively balanced, which helps prevent model bias toward one class.

3.2.2 Text Length Analysis

Text length distribution was analyzed using:

- number of characters
- word count

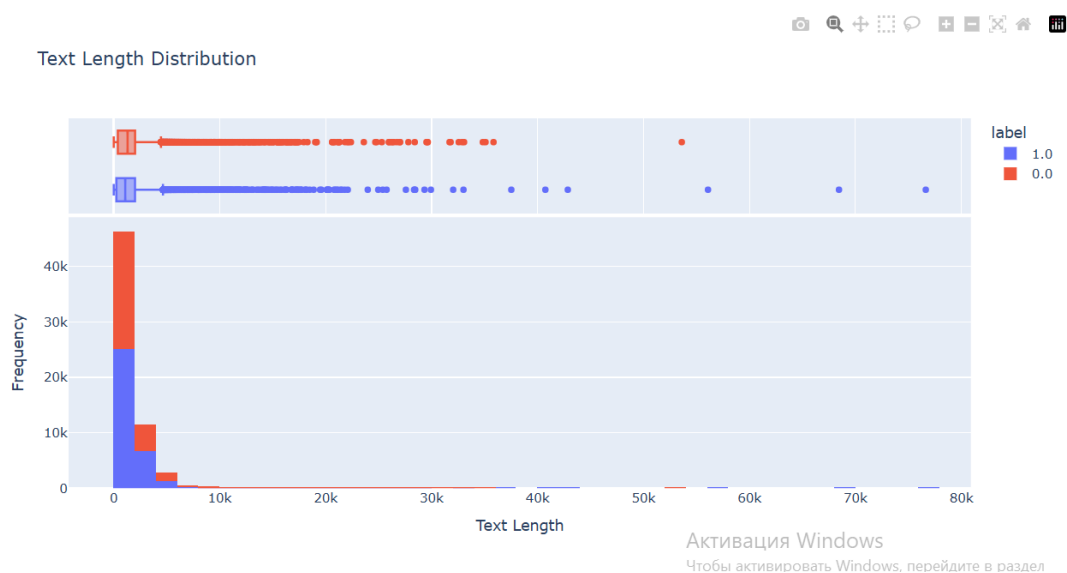


Figure 2: Text Length Distribution

The analysis showed that fake news articles often differ in structure and length compared to real news articles.

3.2.3 Stylometric Analysis

Writing style features such as capitalization and punctuation were analyzed.

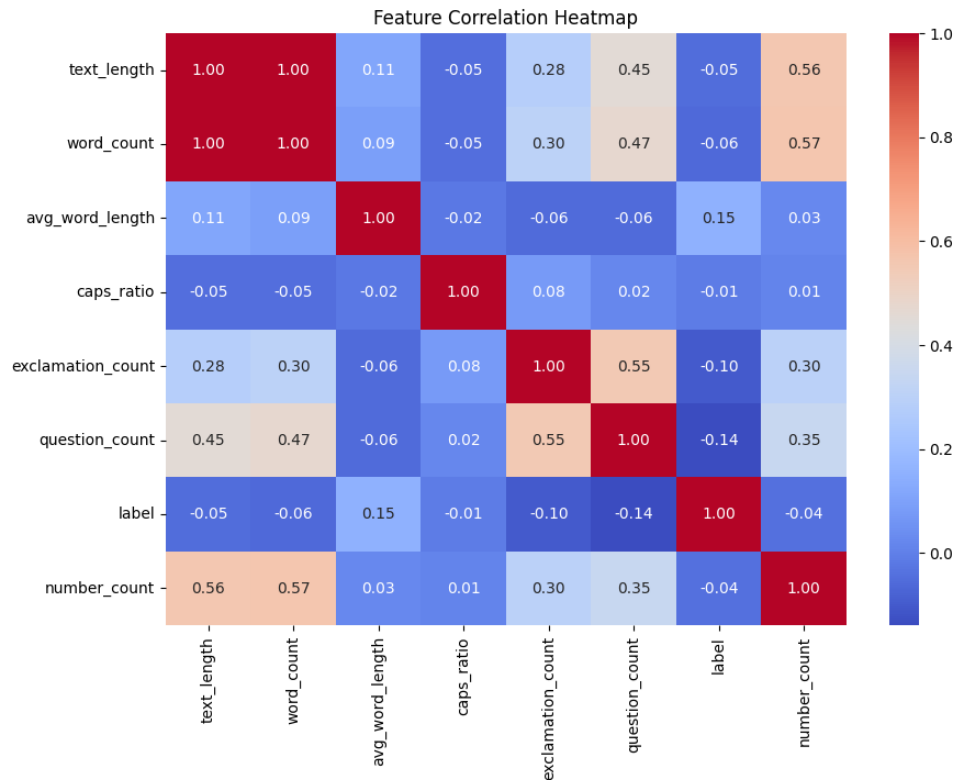


Figure 3: Feature Correlation Heatmap

Key Insights from Correlation Analysis:

- Text length and word count are almost perfectly correlated, meaning longer texts naturally contain more words.
- Longer articles also tend to contain more numerical values.
- Capital letter usage is weakly connected to other textual features, suggesting it does not significantly depend on text length or word count.
- Articles with many question marks often also contain more exclamation marks, indicating expressive or sensational writing styles.
- No single feature alone strongly predicts fake or real news. Therefore, the final prediction relies on a combination of multiple weak signals rather than one dominant feature.

3.2.4 Word Frequency Analysis

The most common words in fake and real news were extracted and visualized.

Word clouds and frequency charts revealed noticeable vocabulary differences between the two classes.

4 Machine Learning Experiments

4.1 Machine Learning Models

Several machine learning models were trained and evaluated:

- Logistic Regression
- Multinomial Naive Bayes
- Random Forest
- Support Vector Machine (SVM)
- XGBoost

Using multiple models allows comparison between different machine learning approaches.

4.2 Train-Test Split

The dataset was divided into:

- 80% training data
- 20% testing data

Stratified sampling was used to preserve class balance.

4.3 TF-IDF Vectorization

Machine learning models cannot directly process raw text.

Therefore, text data was transformed into numerical vectors using:

TF-IDF (Term Frequency – Inverse Document Frequency)

TF-IDF measures how important a word is within a document relative to the entire dataset.

The following configuration was used:

```
TfidfVectorizer(max_features=5000,  
                 stop_words='english')
```

This method converts text into numerical feature vectors suitable for machine learning algorithms.

4.4 Evaluation Metrics

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-score

These metrics are important because fake news detection requires balanced evaluation between false positives and false negatives.

4.5 Model Results

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.819356	0.812747	0.844564	0.828350
Naive Bayes	0.734036	0.756591	0.714542	0.734966
Random Forest	0.816309	0.820189	0.824926	0.822551
SVM	0.821675	0.816511	0.844179	0.830115
XGBoost	0.825782	0.787457	0.907329	0.843154

Table 1: Machine Learning Model Comparison

4.6 Model Comparison Analysis

The experimental results showed that:

- Logistic Regression performed very well, with recall being the highest metric, showing that the model was effective at identifying positive news samples.
- Naive Bayes showed the lowest overall performance among the tested models, indicating limited ability to capture complex textual patterns.
- Random Forest demonstrated stable classification performance with balanced precision and recall
- SVM achieved one of the best overall performances, showing strong capability in handling high-dimensional TF-IDF text features.
- XGBoost achieved the highest recall and F1-score, indicating excellent ability to correctly identify real and fake news articles.

The final model was selected based on the highest F1-score because it balances precision and recall.

4.7 Hyperparameter Tuning

To improve model performance, hyperparameter tuning was performed for the most promising models.

RandomSearchCV was used to search for the best parameter combinations.

The tuning process optimized parameters such as:

- regularization strength
- learning rate
- number of estimators
- kernel parameters
- maximum tree depth

Hyperparameter optimization significantly improved classification performance.

4.7.1 Tuned XGBoost Results

After tuning, the XGBoost model achieved the following performance:

Metric	Score
Accuracy	0.8343
Precision	0.8074
Recall	0.9088
F1-score	0.8551

Table 2: Tuned XGBoost Performance

The tuned XGBoost model achieved very high recall, meaning it successfully identified a large proportion of fake news articles. This is especially important in misinformation detection tasks where missing fake content can have serious consequences.

4.7.2 Tuned SVM Results

Support Vector Machine (SVM) produced one of the best overall performances after hyperparameter tuning.

Metric	Score
Accuracy	0.8598
Precision	0.8633
Recall	0.8786
F1-score	0.8709

Table 3: Tuned SVM Performance

The SVM model demonstrated strong balance across all evaluation metrics. It performed especially well with TF-IDF text features because SVM is highly effective for sparse high-dimensional text data.

The high F1-score indicates strong balance between precision and recall.

4.7.3 Ensemble Model (SVM + XGBoost)

An ensemble model combining SVM and XGBoost predictions was also tested.

The ensemble approach combines strengths from multiple models and often improves generalization performance.

Metric	Score
Accuracy	0.8632
Precision	0.8575
Recall	0.8944
F1-score	0.8755

Table 4: Ensemble Model Performance

The ensemble model achieved the highest overall F1-score among all experiments.

This suggests that combining multiple models helps improve prediction robustness and reduces individual model weaknesses.

4.7.4 Final Model Selection

The final selected model for this project was the Ensemble Model (SVM + XGBoost).

Reasons for selection:

- highest F1-score
- strong balance between precision and recall
- improved robustness
- better generalization ability

The ensemble approach demonstrated the best overall performance for fake news classification.

4.8 Error Analysis

An error analysis step was performed to better understand the strengths and weaknesses of the trained models, by comparing predictions from the SVM and XGBoost models on the test dataset.

The main limitations include:

- sarcasm detection
- context-dependent meaning
- misleading headlines
- highly emotional language

The analysis showed that XGBoost correctly classified a larger portion of disagreement samples compared to SVM.

Based on this observation, a weighted ensemble approach was implemented:

- 60% XGBoost probability
- 40% SVM probability

5 Interactive Demonstration Application

To demonstrate practical usability, an interactive prediction system was created. The application allows users to input a news article, receive prediction results, and classify the article as either Fake or Real.

The demo can be implemented using:

- Jupyter Notebook
- Python script

- Streamlit application

Example workflow:

1. User enters news text
2. Text is cleaned and vectorized
3. Final machine learning model predicts the class
4. Prediction result is displayed

In addition to the classification result, the system provides extended analytical insights, including:

- prediction confidence probability
- sentiment polarity score of the input text
- writing style analysis based on capital letter usage
- text statistics such as word count, sentence count, and character count
- keyword importance visualization using TF-IDF weights
- word cloud representation of the most significant terms

This demonstrates real-world applicability of the developed model.

6 Analysis and Conclusions

6.1 Key Findings

The project demonstrated that machine learning combined with NLP techniques can effectively detect fake news.

Important findings include:

- fake news has distinct linguistic patterns
- TF-IDF is highly effective for text classification
- stylometric features improve classification quality
- SVM and XGBoost performed best

6.2 Limitations

Despite strong performance, several limitations remain:

- sarcasm and humor are difficult to detect
- datasets may still contain bias
- some fake news closely resembles real journalism
- deep contextual understanding is limited

6.3 Future Improvements

Possible future improvements include:

- adding credibility scoring for numerical data to evaluate whether reported statistics are trustworthy or potentially manipulated
- using deep learning models such as BERT or LSTM to contextual meaning in text
- expanding the dataset to include more diverse sources, topics, and writing styles
- multilingual fake news detection
- integrating external fact-checking or verified knowledge sources to validate claims and improve reliability of predictions
- deployment as a web application

6.4 Conclusion

This project successfully developed a machine learning system for fake news detection.

The results show that textual patterns and writing style differences are strong indicators of misinformation. Machine learning models achieved high classification performance and demonstrated practical usability through an interactive prediction system.

Overall, the project confirms that machine learning can be an effective tool for combating misinformation in digital environments.

References

Datasets

- ISOT Fake News Dataset
Available at: <https://www.kaggle.com/datasets/clmentbisailon/fake-and-real-news-dataset>
- LIAR Dataset
Available at: <https://github.com/chengxuphd/liar2/tree/main/liar2>
- WELFake Dataset
Available at: <https://www.kaggle.com/datasets/saurabhshahane/fake-news-classification>

Research Papers

1. Shu K., Sliva A., Wang S., Tang J., Liu H. *Fake News Detection on Social Media: A Data Mining Perspective*. SIGKDD Explorations, 2017.
2. Jurafsky D., Martin J. H. *Speech and Language Processing*. Stanford University, 2023.
3. Lazer D. M. J. et al. *The Science of Fake News*. Science, Vol. 359, Issue 6380, pp. 1094–1096, 2018.